*Machine Learning A*

*2026-2027*

**Preparatory Exercises 2**

**Sadegh Talebi** **Yevgeny Seldin**

Department of Computer Science
University of Copenhagen

# 1 Confidence Intervals for Bernoulli

We are given a sequence $S = (X_t)_{1 \le t \le n}$, which is drawn i.i.d. from a Bernoulli with mean $\mu$; the sequence is collected in `S.csv`. Compute 0.95-CI (two-sided) for $\mu$ using Hoeffding's inequality.

*Deliverables:* A confidence interval with computation details. Final answer without explanation will earn zero point.

# 2 Hoeffding's Inequality: A Warm-up Example

Suppose we have a standard six-sided die. Consider 80 independent throw of the die, and let $S$ denote the sum of the realized numbers after $n$ throw. Let $p$ denote the probability that $S > 320$. Derive upper bounds on $p$ using Hoeffding's inequality.

*Hint: When using Hoeffding's inequality, pay attention to the range of the random variable in question.*

# 3 Comparison of Markov's, Chebyshev's, and Hoeffding's Inequalities

Suppose we have a standard six-sided die. Consider $n$ independent throw of the die, and let $p$ denote the probability of observing a '6' for at least $n/4$ times.

More precisely, define $X$ to denote the number of times a '6' is observed in this experiment. Then, $p = \mathbb{P}(X \geq n/4)$.

Derive upper bounds on $p$ using Markov's, Chebyshev's, and Hoeffding's inequalities. Compare the three upper bounds. Include computation details; final answers without explanation will earn zero point.

*Hint: When using Hoeffding's inequality, pay attention to the range of the random variable in question.*

# 4 Combining Multiple Confidence Intervals

Alex works at a pharmaceutical company. She has developed a new treatment and is interested in estimating its success probability $\mu$. To this end, she has collected an i.i.d. sample $S$, from which she computed the sample mean $\widehat{\mu}$, as an estimate of $\mu$. However, the company requires a confidence interval (CI) for $\mu$ too.

Rather than computing the CI herself, she decided to outsource the task to three data science companies, who are world-class experts in this area and have developed advanced tools to compute confidence intervals. The interaction protocol with each company $i \in \{1, 2, 3\}$ is as follows:

- The company $i$ receives the dataset $S$ as well as a parameter $\delta_i$ as input, and returns $\texttt{CI}_i = [l_i, u_i]$ that is a $(1-\delta_i)$-CI for the mean of the distribution underlying $S$.
- The company does not reveal how they build $\texttt{CI}_i$, and hence the 3 companies may use different methods.

(For example, the companies could use different concentration inequalities to construct their CIs.)

Assume that the project requires a 0.99-CI on $\mu$. Explain how Alex could choose $\delta_i, i = 1, \ldots, 3$ when contracting each company and how she must use the CIs received from the companies (i.e., what is the best CI she can report to her boss).